1.8.1 Properties of cubes of numbers

S. No.	Properties	Examples
1.	The cube of a positive number is positive.	$6^3 = 6\times6\times6 = 216.$
2.	The cube of a negative number is negative.	$(-7)^3 = (-7)\times(-7)\times(-7) = -343$
3.	The cube of every even number is even.	$8^3 = 8 \times 8 \times 8 = 512$ is even
4.	The cube of every odd number is odd.	$9^3 = 9 \times 9 \times 9 = 729$ is odd
5.	If a natural number ends with 0, 1, 4, 5, 6 or 9, its cube also ends with the same 0,1, 4, 5, 6 or 9 respectively.	$10^3 = 1000$, $11^3 = 1331$, $14^3 = 2744$ $15^3 = 3375$, $16^3 = 4096$, $19^3 = 6859$
6.	If a natural number ends with 2 or 8, its cube ends with 8 or 2 respectively.	$12^3 = 1728$, $18^3 = 5832$
7.	If a natural number ends with 3 or 7, its cube ends with 7 or 3 respectively.	$13^3 = 2197$, $17^3 = 4913$
8.	The sum of the cubes of first n natural numbers is equal to the square of their sum.	$1^3 + 2^3 + 3^3 + ... + n^3 = (1+2+3+.....+n)^2$ Check that, $1^3 + 2^3 + 3^3 + 4^3 = (1+2+3+4)^2$

Note

- A perfect cube does not end with two zeroes.
- The cube of a two digit number may have 4 or 5 or 6 digits in it.

Try these

Find the ones digit in the cubes of each of the following numbers.

1. 12 2. 27 3. 38
4. 53 5. 71 6. 84

1.8.2 Cube root

The cube root of a number is the value that when cubed gives the original number.
For example, the cube root of 27 is 3 because when 3 is cubed we get 27.

Notation:

The cube root of a number x is denoted as

$$\sqrt[3]{x} \text{ (or) } x^{\frac{1}{3}}.$$

Here are some more cubes and cube roots:

$\sqrt[3]{1} = 1$ since $1^3 = 1$, $\sqrt[3]{8} = 2$ since $2^3 = 8$,
$\sqrt[3]{27} = 3$ since $3^3 = 27$, $\sqrt[3]{64} = 4$ since $4^3 = 64$,
$\sqrt[3]{125} = 5$ since $5^3 = 125$ and so on.

Cubes	Cube Roots	Cubes	Cube Roots
1	1	729	9
8	2	1000	10
27	3	1331	11
64	4	1728	12
125	5	2197	13
216	6	2744	14
343	7	3375	15
512	8	4096	16

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